

The Fordist Revolution: Social Reproduction, Commodity Chains, and the Global Division of Labor

Introduction

Fordism has been a slippery concept within historical writing over the last 100 years. Although labor historians and economists agree that it is a central moment in both American and global economic history, there is sometimes little agreement on what Fordism actually *is*. In 1990, in a talk humorously titled “What in the F—’s name is Fordism”, British sociologist Simon Clarke noted the imprecision in discussions of Fordism and whatever non-Fordist capitalist epoch is supposed to be its successor: “Much energy has been expended in debating the diffuse characterizations of ‘Not-Fordism’. However, much less attention has been paid to the characterisation of Fordism.”¹ Even the popular legacy of Henry Ford himself is deeply ambivalent. He was a benefactor who improved the lives of his workers and a tyrant whose Sociological Department spied on their personal lives; he created the modern middle class but also initiated a culture of consumption that is currently eating the planet; he was a humanitarian who founded schools for the his mostly immigrant employees and he was a vociferous anti-semite. Only one thing is uniformly agreed upon: Fordism represented a genuine break with existing patterns of production in capitalist society; it was a revolution.

The boundaries of this revolution are blurred. Daniel Watson identifies two main approaches to Fordism: a narrow view of Fordism as a system of mass production centered around Henry Ford’s innovative assembly line, and a broader view of Fordism as a “phase of capitalist development taking place several decades after the heyday of the Ford Motor

¹Clarke, Simon. 1990. “What in the F—’s name is Fordism.” Paper presented at British Sociological Association Conference, University of Surrey

Company” that is characterized by a “careful balance in the american economy between mass production and mass consumption and an increase in the wage rate and social provisions for workers”.² In this broader view, the Fordist era was initiated by Henry Ford’s productive innovations, but was not fully realized until decades later. I am partial to this latter interpretation of Fordism, and I see Fordism as a global phenomenon, but I also appreciate the specificity of narrowly grounding the Fordist break in the early automotive industry in Michigan. It is not true that mass consumption, rationalization of labor, and social provision for workers were truly exported around the world. Rather, the social gains granted to (some) workers were propped up by a deepening of labor stratification both locally and globally.

Writing in 1982 Alain Lipietz, characterizes the rise of Fordism as an object of economic history, noting that

in three ‘glorious’ post-war decades, a crucial change was introduced by the system of capitalist accumulation that various French economists, following Gramsci, have baptized ‘Fordism.’ A ‘centre’, displaying a cohesive internal logic at the level of blocs (North America, the EEC) if not individual countries, came to stand against a ‘periphery’ which, whatever the internal dynamic of its component parts, was ever more disconnected from this ‘autocentred’ logic.³

However Lipietz immediately questions whether we can even “speak of a world-wide extension of Fordism,” noting that during the late 1960s and early 1970s, the economic periphery experienced uninterrupted growth while the economic core went into a production crisis. Indeed, the industrialization of the periphery was not a slow march in the footsteps of the core, but was its own specific development, conditioned by and conditioning the dynamics of accumulation in the United States and Europe. Although Lipietz was writing about the mid-twentieth century, his questioning of neat divisions between the core and periphery, and of stagist narratives of Fordism are relevant to the global dynamics which were already present in the early automotive

²Watson, Daniel. 2018. “Fordism: A Review Essay.” *Labor History*. 2

³Alain Lipietz. 1982. “Towards Global Fordism?” *New Left Review* I:132. 33-47.

industry, decades before anybody considered Fordism to be a global phenomenon.

Neat narratives about the Fordist worker and the family wage do not hold up to scrutiny. In Michigan's auto factories in the first decades of the twentieth century, local workers were increasingly divided into two camps. On the one hand, manufacturing tasks were decomposed into minimal parts, rationalized to the extreme and voided of intellectual stimulation. On the other hand, a great class of managers and engineers was cultivated to design the machines and processes that would combine the decomposed labor of others into finished products - following a managerial revolution that Alfred D. Chandler called the 'visible hand' of the market.⁴ Outside the factories, the living conditions of workers were similarly divided. Skilled and semi-skilled workers, mostly white, were inducted into a class of homeowners, while racialized and menial laborers often lived in crowded slums.

Of course, the labor that contributed to the manufacturing of automobiles was not confined to factory work. Raw materials had to be imported - especially rubber which needed to be sourced from the global south. The stratified labor market in industrial Michigan was mirrored by a global division of labor, in which wild rubber harvesters in the Amazon and the Congo, and to a lesser extent the later rubber plantation workers in southeast Asia, occupied an intensely degraded position. The labor conditions were harsh, and so were the living conditions.

This paper takes up early labor stratification in the automotive industry and the methods used to create and discipline it. Keeping the complications listed above in mind, I hope to situate early Fordist production within the birth of what Giovanni Arrighi called the "US systemic cycle of accumulation" - the period of historical capitalism covering much of the twentieth century when the United States dominated global capitalist accumulation. According to Arrighi, this

⁴Alfred D. Chandler. 1977. *The Visible Hand: The Managerial Revolution in American Business*. Cambridge: Harvard University Press.

cycle of accumulation was distinguished by “an internalization of transaction costs,”⁵ in which firms vertically integrated in order to retain the surplus value otherwise lost through the purchase of inputs. The scope of this integration was explicitly global, as accumulation in the core has historically been dependent on the exploitation of peripheral regions, and as the working conditions in the periphery responded to demands of accumulation in the core. However, the methods of production and labor control closely related to Fordism, such as assembly lines, the decomposition of tasks, and a family wage, were *not* generalized globally. This raises the question of how vertical integration was effected across national lines, or perhaps more accurately, what were the effects of Fordist production practices in the core on labor conditions in the periphery?

The second purpose of this paper is to look at social reproduction within this global context - and especially to highlight the centrality of housing to Fordist social policy. Linking these two concepts - the global scope of early Fordism and its regime of social reproduction - is the commodity chain. Wallerstein and Hopkins used the term commodity chain to conceptualize the interconnectedness of global capitalism: “take an ultimate consumable item and trace back the set of inputs that culminated in this item - the prior transformations, the raw materials, the transportation mechanisms, the labor input into each of the material processes, the food inputs into the labor.”⁶ The commodity chain is the axis along which Fordist firms tried to integrate, but this chain includes more than just raw materials and the labor that works them: it is imbricated with “the reproduction of the labor forces involved in these productive activities.”⁷ This paper aims to look at the configuration of working class housing as a component of social reproduction

⁵Giovanni Arrighi. 2009. *The Long Twentieth Century: Money, Power, and the Origins of Our Time*, London and New York: Verso Books, 247.

⁶Terence K. Hopkins and Immanuel Wallerstein. 1977. “Patterns of Development of the Modern World-System,” *Review* 1:2. 128.

⁷Hopkins and Wallerstein, “Patterns of Development,” 128

along the global Fordist commodity chains. The scope of this paper will focus primarily on the origins of Fordism: the production of automobiles in the first decades of the twentieth century. By clarifying the stakes of early Fordism as a global project, and housing as one of its crucial tools, I hope to prepare the ground for a broader investigation of the manipulation of working-class housing as constituent of the capitalist world-system.

Fordism

It would be controversial to identify ‘Fordism’ as such solely with the practices of the Ford Motor Company and its competitors in the auto industry. Certainly there is a relationship between Ford and Fordism, but they are not identical. And if we take the ‘world of production’ to mean all production practices across the globe, then characterizing it as ‘Fordist’ is certainly untenable - although it may be dominated by Fordism in a certain sense. For the time being, let us set aside the question of the ‘world of production’ and try to clarify just what is meant by the word ‘Fordism’.

In the popular imagination, Fordism comprises two moments of personal genius: Henry Ford’s implementation of the automated assembly line and his institution of a family wage with the Five Dollar Day. Ford notoriously combined these principles in his River Rouge factory, a single complex that took raw inputs and used well-paid, middle class laborers to turn them into cars. His win-win solution to the antagonism of labor and capital transformed American life, and facilitated the dominance of US industry on the world stage. He combined innovation on the shop floor with a reconfiguration of family life, ushering in the modern middle class and an era of mass consumerism.

This popular conception is problematic on several counts. Ford’s production practices

were genuinely innovative, but they were often adopted and improved upon by his competitors - notably General Motors. Ford's most famous social policy, the Five Dollar Day, was not a universal wage increase, but a profit share given only to employees whose private lives were approved by agents of the Ford Sociological Association. Further, although both of these practices were implemented by the Ford Motor Company before the first world war, they were not generalized on a national scale until the 1940s.

Michel Aglietta provides us with a classic structuralist account of Fordism as a generalized set of production principles, historicizing it within the long cycles of historical capitalism. For Aglietta, although Fordist practices may have originated in the auto industry, Fordism represents a societal-level shift in the relations of production and the reproduction of capital. In *A Theory of Capitalist Regulation*,⁸ he criticizes the dominant economic theories of capitalist equilibrium for being, among other things, ahistorical and abstract. Against this, Aglietta denies "that what exists does so automatically", necessitating a theory of social reproduction, and he claims that "when actual social systems are studied, historical experience confirms that *transformation means rupture, qualitative change*."⁹ He attempts to situate the Fordist moment within the turbulent history of capitalist production.

Taking the US since 1870 as his spatial-temporal object, Aglietta elaborates a theory of capitalist regulation as an attempt to "formulate the general laws the way in which the determinant structure of a society is reproduced."¹⁰ In the decades after the American civil war, these laws were oriented towards the development of heavy industry. Claiming that "the fundamental economic condition for industrial capitalism to develop is the formation of a growing agricultural surplus product and its realization in commodity form," Aglietta highlights

⁸Aglietta, Michel. 1979. *A Theory of Capitalist Regulation*. London: NLB

⁹Aglietta, *Capitalist Regulation*, 12

¹⁰Aglietta, *Capitalist Regulation*, 13

19th century America's privileged situation. The incorporation of vast frontiers precipitated the development of rail infrastructure, irrigation, farming and mining. This 'regime of extensive accumulation' was funded and aided by the development of western urban centers and their growing financial powers.

In the first decades of the twentieth century, this regime fell into crisis. Aglietta positions the economic collapse of the 1930s as a result of overaccumulation and underconsumption. Specifically, the average worker was paid so little that they could not contribute to the valorization cycle by purchasing and consuming commodities. As capital entered a crisis in the 1930s, producers sought out new ways to maintain profitability. Fordism, as the model set out by the Michigan automotive industry, was their answer to changing economic conditions.

Fordism figures into Aglietta's account as an 'intensive regime' which "creates a new mode of life for the wage-earning class by establishing a logic that operates on the totality of time and space occupied or traversed by its individuals in daily life."¹¹ In other words, Fordism is characterized by an unprecedented expansion of capitalist interest and control over the lives of its waged workers beyond the shop floor. Aglietta terms this reorganization of social life a 'mode of consumption' which is oriented towards the renewal of the energy expended in labor processes and the maintenance of a continual supply of labor power.

As an articulation between production and consumption, Fordism comprised several specific innovations linked to more general social reconfigurations. On the production side, Aglietta identifies the automated assembly line as Fordism's "characteristic labour process."¹² The assembly line separated mental and manual labor to a previously unseen degree. Instead of managing the relationship between several related tasks, workers performed simplified rote tasks

¹¹Aglietta, *Capitalist Regulation*, 71

¹²Aglietta, *Capitalist Regulation*, 117

over and over: “the labour process [was] converted from a dense network of relationships between jobs, with intermediate products passing back and forth ... into a straightforward linear flow of the material under transformation.”¹³

The mental component of labor was excised from the ambit of the general laborer and given to a growing class of managers and process engineers. Workers became fixed to clearly and rigidly defined roles that were “determined by the configuration of the machine system”¹⁴ which was “planned in engineering departments which were themselves subject to the division of labor.”¹⁵ The Fordist workplace began to look more and more like that described by Marx: living labor ruled by the dead labor that was embodied in the planning and manufacturing of the machine system.

This process of rationalization extended far beyond the shop floor. Along with the revolution in production, Fordism reconfigured the social and private lives of the workforce. Aglietta notes that “for the first time in history, Fordism created a norm of working-class consumption in which the general ownership of commodities governed the concrete practices of consumption.” Two commodities stand out as governing this norm: the automobile as a means of private transportation, and standardized housing. Both of these commodities transformed the urban spaces in which the working classes were renewed, and their relatively high cost “presupposed a vast socialization of finance, and correlatively a very strict control over workers’ resources and expenditures.”¹⁶ However, Aglietta notes that the rise of certain family norms within a consuming class of workers was never quite general: Fordism also deepened the stratification of the working classes, which “forms part of the concrete conditions of reproducing

¹³Aglietta, *Capitalist Regulation*, 118

¹⁴Aglietta, *Capitalist Regulation*, 118

¹⁵Aglietta, *Capitalist Regulation*, 119

¹⁶Aglietta, *Capitalist Regulation*, 159

the wage relation.”¹⁷

Aglietta’s account of Fordism as an historical articulation of means of production and consumption responding to the global needs of capital accumulation is a compelling one, but it is not without its critics. In 1988, Simon Clarke published a critique of Aglietta and the Regulationist school, whose approach he deems overly structuralist. Clarke claims that for the Regulationists, “in the relation between structure and struggle, it is the class struggle which takes the subordinate role, its significance constrained by the structural imperatives of the regime of accumulation.”¹⁸ On the question of working-class struggle this charge is perhaps more applicable to Aglietta’s followers than to his own work, which deals at length with working class organization. But class struggle is not one-sided, and Aglietta fails to adequately theorize the role of the state in constituting regimes of capitalist regulation.

This calls into question the framework of extensive vs intensive regimes of accumulation, or at least Aglietta’s expression of it. Clarke claims that “the post-war boom was not initiated by Fordist modes of regulation ... but by austerity, by sharp industrial and political class struggles, and by direct state intervention to establish both a high rate of profit and the proportionality of the various branches of production at the national and international levels.”¹⁹ The economic policy of Keynesianism was not an immanent development of the structure of capitalist regulation, but was dictated by “the balance of class forces, institutionalised in the systems of industrial relations, electoral politics and the welfare state.”²⁰ In short, the regulation approach “overestimates the stability and duration of ‘Fordist’ modes of regulation” and “the extent to which Fordism can be considered a systematic ‘regime of accumulation’”²¹

¹⁷Aglietta, *Capitalist Regulation*, 137

¹⁸Clarke, Simon. 1988. “Overaccumulation, Class Struggle and the Regulation Approach.” *Capital & Class*. Vol 12:3.

¹⁹Clarke, “Overaccumulation” 14-15

²⁰Clarke, “Overaccumulation” 18

²¹Clarke, “Overaccumulation” 16

A related failing of Aglietta's work emerges from his (non)treatment of imperialism. In the introduction to *A Theory of Capitalist Regulation*, he defines imperialism as a system "through which one state manages to influence a series of other states to adopt a set of rules that are favourable to the stability of a vast space of multilateral commodity relations guaranteeing the circulation of capital."²² After posing this anemic definition, he pushes it aside, claiming that "there could be little question of analysing in the present work the system of contemporary states and the transformation that the crises of the 20th century have produced in inter-state relations."²³ But he is unable to escape the weakness of his conception of capitalist hegemony - it undermines his entire project. In his attempt to isolate America from the global south, he misapprehends the extent to which American industry was dependent on an unequal global division of labor. The unequal exchange across state lines undergirds and forms part of capitalist regulation. The definition of imperialism as management inflicted by one state on several with the goal of broadening the circulation of capital prevents Aglietta from appreciating the global scope of capitalist regulation and reproduction.

If we have called into question Fordism as a systematic regime of accumulation, and its validity as a mode of regulation divorced from both state action and the global contours of capital, we might reasonably ask what use the concept of Fordism provides, and what value we can gain from Aglietta's treatment of it. Two of Aglietta's core interventions - Fordism as a specific qualitative change in the articulation of production and consumption, and the increased stratification of the wage-earning class - are crucial and worth preserving, but we must investigate the extent to which the articulation in American industry was generalized beyond US borders. Specifically, we must examine the effect of a reconfiguration of US production on

²²Aglietta, *Capitalist Regulation*, 32

²³Aglietta, *Capitalist Regulation*, 32

capitalist regulation in other parts of the globe. In the following sections, we will look at Fordism both in its origins in the Michigan automobile industry, and its application beyond US borders.

Local Fordism: the Labor History of Fordist Michigan

As the birthplace of Fordism, industrial Michigan bears the imprint of Fordism's early development and its essential dynamics. Since the Second World War, American labor historians have been arguing over these dynamics. Marxist historians in the US emphasized the labor process itself as an essential component of labor history. Harry Braverman's *Labor and Monopoly Capital: The Degradation of Work in the Twentieth Century* centered the deskilling of the workforce as a method of labor control. Nelson Lichtenstein writes in the introduction to *On the Line: Essays in the History of Auto Work* that "Braverman broke new ground by alerting historians and sociologists to the social context within which the technology of production was embedded."²⁴

Braverman's view was oversimplified, however, and labor historians debated "the extent to which craft skills were actually eroded in mass production industry."²⁵ Rather, the labor market was bifurcated between low-paid and low-skilled workers and highly paid engineers and managers. The partial deskilling of the workforce was a point of tension between labor and capital. At times capitalists strategically degraded the skill of workers in order to disempower them, while the unions resisted this degradation.

It is often repeated that unions in late 19th, early 20th century Michigan were weak and listless - supposedly this contributed to the rise of Fordism as a new method of labor exploitation. However, this claim is challenged by Thomas Klug, in an essay in *On the Line*. As

²⁴Nelson Lichtenstein. 1989. "Introduction," in *On the Line: Essays in the History of Auto Work*, eds. Nelson Lichtenstein and Stephen Meyer. Chicago: University of Illinois Press. 5.

²⁵Lichtenstein, "Introduction," 5.

the lumber industry declined at the close of the century, economic activity in the state shifted towards the growing cities. In the 1890s, Detroit was the stove producing capital of the world, with a significant presence of other types of metalwork that would be crucial to the development of the auto industry. Most of the organizations representing workers were craft unions, and they admitted only skilled labor to their ranks. However, the assertion that these unions were powerless is misguided. Klug notes that “although only a few employers of iron molders formally agreed to the exclusive employment of union men, many foundries were de facto closed shops.”²⁶ The reason for this is the Iron Molders Union’s strategy of organizing the “best molders” so that any employer wanting good labor had to go through the union.

The craft unions in fact posed enough of a threat to employers that in 1902, “metal trade employers organized the Employers’ Association of Detroit (EAD).”²⁷ They felt that “the brief accommodation with labor unions in the late 1890s and early 1900s had gained them few benefits Rather, strikes became more frequent, and unions gained additional recruits.”²⁸ The unions had developed significant leverage in eastern Michigan and prevented employers from undermining labor rights to increase productivity and profitability. As a unified bloc, the EAD was able to enforce what they considered to be their right to pay workers based on the quality of their individual work, and to hire from outside the local labor pool.

Detroit employers were also troubled by what they saw as a decrease in control over the labor process itself. As the unions had a monopoly on skilled labor, which was crucial to the operation of heavy manufacturing. John J. Whirl, a member of the EAD, said that “in some of our industries the proprietors were merely the financial agents to purchase the supplies, pay the wages, and sell the manufactured products ... As to the operations of the factory, they had no

²⁶Thomas Klug. 1989. “Employers’ Strategies in the Detroit Labor Market, 1900-1929,” in *On the Line: Essays in the History of Auto Work*, eds. Nelson Lichtenstein and Stephen Meyer. Chicago: University of Illinois Press. 44.

²⁷Klug, “Employers’ Strategies,” 48.

²⁸Klug, “Employers’ Strategies,” 48.

more real control than if they were in no way connected with it.”²⁹ The attempt by employers to band together and undermine the craft unions was joined with an understanding that to truly gain control of their businesses, they would need to reduce their reliance on a small pool of skilled labor: an understanding that would animate the coming Fordist revolution.

And it was a revolution. The automobile was not invented in Michigan – in fact the first two viable gasoline powered cars were built in Massachusetts and Indiana in 1892 and 1894.³⁰ But Michigan arrived early on the automotive scene, propelled by easy access to machining, raw materials, and capital: in 1896 Ransom E. Olds of Lansing and Detroit’s Charles B. King and Henry Ford demonstrated viable motor powered carriages. By the early twentieth century, there were dozens of auto companies in Michigan, but they faced significant issues.

Initially, auto production was a craft industry: each car was handmade by skilled craftsmen. Due to the competitive market in Michigan, it was difficult for employers to keep workers from leaving for greener pastures at another company. Likewise, the cars were expensive, and so there was a limited market for the product being produced. Henry Ford’s solutions to these problems transformed the industry.

In 1908, Ford introduced the Model T, a mass produced car with standardized parts that required less skill to make and cost far less than competitor’s cars, contributing to a rise in sales. In 1913 he introduced the automated assembly line, further standardizing production and reducing costs. And in 1914, he introduced the Five Dollar Day, a wage increase that solved the problems of employee turnover and bloated training costs. In order to compete, rival firms had to adopt and advance Fordist innovations and in the 1920s, General Motors introduced the yearly model change, further driving up consumer demand.

²⁹Klug, “Employers’ Strategies,” 49.

³⁰Dunbar, *Michigan: A History of the Wolverine State*, 552.

The challenges that Fordism introduced to the labor market and to the shop floor have been of particular interest to labor historians working in the Marxist tradition. From the early suppression of union activity by auto companies to the 1936 sit-down strike in Flint and the rise of the most militant unions in American history, the Fordist factory is a towering edifice in American labor historiography. But a full accounting of the social change wrought by Fordism would require a labor history that is critical of the notion of labor itself.

Liberal and Marxist historians in the twentieth century disagreed about the political nature of their work and the goal of the labor movement, but there is a fundamental similarity in their analysis. The key category for both traditions is labor, but while labor historians interrogated its relations with other social forms such as capital or law, they were less critical of the category of labor itself. Labor and by extension laborers are taken to be self-evidently coherent classifications: generally wage-laborers, generally men, generally working for someone else.

In 1971, Thomas Krueger, in “American Labor Historiography,” narrates a partial break with the shared paradigm of the liberal and Marxist historiography, but at the time of his writing that break was still incomplete. He is concerned with a shift away from Common’s conception of labor history, citing Hyman Berman’s opinion that labor history “is more than the rise or fall or the failures and successes of organizations” and David Montgomery’s call to research “the cultural and intellectual life of the working classes.”³¹ This new direction follows the work of the British Marxists in expanding the category of the working classes to include a greater sample of the relations of working class life, including community support of labor against capital during strikes, the influence of religion, and the permeability of class categories.

³¹Berman and Montgomery quoted in Thomas Krueger. 1971 “American Labor Historiography, Old and New,” *Journal of Social History* 4:3. 281-2

Krueger notes that in 1971, this development was still immature, and had to “solve some of its intellectual problems.”³² His framing of the task ahead, however, demonstrates just how partial the break with old labor history still was. He lists ten “imposing questions” that need to be answered by new labor historians, and while they demonstrate an expansive view of labor *history*, they hew to a decidedly traditional view of the category of labor itself. One question is representative: “What happened to blacklisted workers, to men over 40 who got fired, to families whose head was mangled or killed on the job.”³³ Implicit in this question is the orientation of a nuclear family around labor and the elevation of specifically male waged labor.

In the 1970s, a significant challenge was posed to this traditional notion of labor, and to both Marxist and classical labor historiography, by feminist movements such as the Wages for Housework campaign. Though not a primarily American movement, the challenge posed by radical and Marxist feminists fundamentally changed the way that labor history in the US could be written. Writers and activists in this tradition challenged the notion that only waged labor contributed to the economy and took traditional Marxism to task for accepting the capitalist devaluation of feminized work.

Sylvia Federici, in the 1975 essay “Counterplanning from the Kitchen” lays out the feminist challenge to a wage-centric notion of labor posed by Wages for Housework:

Beginning with ourselves as women we realize that working for capital does not necessarily produce a paycheck and does not begin and end at the factory gates.... Housework, in fact, is much more than housecleaning. It is servicing the waged workers physically, emotionally, and sexually and getting them ready to work day after day for the wage.... This means that behind every factory, every school, every office, and every mine there is the hidden work of millions of women, who consume their life reproducing those who work in those factories, schools, offices, and mines.³⁴

The family life of workers, in other words, is not a realm separate and apart from their

³²Krueger, “American Labor,” 283.

³³Krueger, “American Labor,” 284.

³⁴Sylvia Federici, 2021. *Patriarchy of the wage: Notes on Marx, Gender, and Feminism*. Oakland: PM Press, 2021. 11-12.

labor but is constitutive of it. The unpaid work of women (and children) forms the base on which the institution of waged labor stands. Implicit in Federici's quote (and explicit in her other work) is the accusation that labor history and labor activism, by ignoring the work of women, reproduces capital's own devaluation of gendered and non-waged work.

The point of this intervention is not to merely add a new type of labor to an old conception: the new category is not labor plus gender. A labor history that merely traces struggles and changes in the structure of housework would do little more than transplant the old framework of analysis onto a new subject. Likewise, a treatment of gender as a variable that affects only women would be counterproductive. Rather, the task for feminist labor history is to interrogate the very process of categorization. Elizabeth Faue, introducing a volume of essays on gender and labor history, claims that "labor historians' use of collectivities as units of analysis – whether classes, communities, trades, or families – masked both intraclass division and an understanding of conflicting interests that was at times one-dimensional."³⁵ To remedy this, we must examine not how the category of gender *acts on* the category of labor, but how it permeates and *constitutes* it.

Two complementary theoretical traditions are of use to us here. Social Reproduction Theory and World-Systems Analysis employ the overlapping categories of the family (SRT) and the household (WSA) to examine the gendered dynamics of labor. Labor is reproduced in different ways for different circumstances/locations and the category of labor itself has been opened up to include processes outside the shop floor. How are these processes disciplined and managed?

For Immanuel Wallerstein, the household is a crucial element of the capitalist world-

³⁵Elizabeth Faue. 1993. "Gender and the Reconstruction of Labor History: An Introduction," *Labor History* 34:2-3. 171.

system. Borrowing from Marxist thought, he identifies proletarianization as one of the constitutive processes of historical capitalism, but contra orthodox Marxism, he associates it not with an individual worker, but with households. Households are “relatively stable structures which share a common fund of current income and accumulated capital”³⁶ – in other words they are groups of people who pool their income from various sources in order to meet the collective needs of survival. This income (broadly conceived) can come from waged work, petty commodity production, unwaged reproductive work, or even cash transfers from kin or government agencies.

Social Reproduction Theory hews more closely to Marxist feminist categories. Though SRT theorists disagree on exact demarcation, there is a general distinction between productive labor (generally that which is done for a wage) and unproductive/reproductive labor (that which is concerned with providing for the base needs of life, on either a waged or unwaged basis). As Nancy Fraser puts it, the unwaged work of (usually) women supports the renewal of the life of the waged worker and forms “one of [the] background conditions of possibility” of capitalist accumulation.³⁷

Both of these frameworks can grant us insight into the management of labor in Detroit around the turn of the century. Prior to 1914, when Henry Ford instituted the famous Five Dollar Day in an attempt to reduce employee turnover, it was uncommon for Michigan families to live on the wage of a single earner. Working-class families cobbled together livings from various sources, fighting both to make higher wages and to carve out more time to do necessary subsistence labor. In “The Detroit Story: the crucible of Fordism,” Kathleen Stanley and Joan Smith examine the extent to which working-class families were transformed by the family-wage

³⁶Immanuel Wallerstein. 1983. *Historical Capitalism*. London: Verso Editions. 23.

³⁷Nancy Fraser. 2017. “Crisis of Care? On the Social-Reproductive Contradictions of Contemporary Capitalism” in *Social Reproduction Theory: Remapping Class, Recentring Oppression*. ed. Tithi Bhattacharya. London: Pluto Press. 23.

in Detroit from roughly 1890 through the middle of the twentieth century. They build on the category of the household as an income-pooling unit from world-systems analysis and take the family as a site of wage-contestation. Working-class households piece together subsistence from various sources: beyond the wages of a single laborer, they rely on the unpaid subsistence labor of women and children, charity or community assistance, income from boarders, informal commodity production, and many other sources. The need to depend on these extra-wage sources of income would be alleviated with a family-wage.

Smith and Stanley point out that in the 1890s, wages in the Detroit area were much lower than the national average. This was partly due to the depressed wages of immigrants who made up close to 70% of the city, and due to the relatively large percentage of the workforce that were women. While noting that there was much variation among family incomes, they cite that “The [average] laborer whose budget was reported by the Michigan Bureau of Labor, for example, faced an annual shortfall of \$86, or 27 percent of his total income.”³⁸ Over a third of the city’s workforce was made up of “semiskilled, unskilled, and service workers” who earned considerably less than this average.

In addition to the type of labor performed, a major factor in the ability of Detroit workers in 1890 to survive on a wage was family composition, which correlated strongly with race. Polish immigrants generally had the largest households of any ethnic group, much greater than American-born whites. While most of the households across all ethnic groups were organized as nuclear families, Black families were far more likely to either not have children, or to have only one parent. Stanley and Smith note that while smaller families are generally considered to be more sustainable in times of depression (such as the 1890s), the significantly larger-than-average

³⁸Kathleen Stanley and Joan Smith. 1992. “The Detroit Story: the crucible of Fordism,” in *Creating and Transforming Households: the constraints of the world-economy*, eds. Joan Smith and Immanuel Wallerstein. Cambridge University Press. 36.

Polish households were able to leverage their family composition into more effective “patterns of subsistence.”³⁹

It appears that “Polish families brought with them skills associated with their recent past as peasants, which could be put to use outside the wage nexus.”⁴⁰ While it was uncommon for married Polish women to work outside the home for a wage, they were more likely than other groups to maintain livestock and vegetable gardens. When cash flow was an issue, Polish children were sent to work at higher rates than children of other families. The various forms of subsistence labor and multiple sources of wage income combined to insulate Polish families from both economic downturns and a reliance on a single wage-earner.

Other ethnic groups in the 1890s had their own strategies for survival that combined wage and non-wage incomes. Black families in Detroit generally had fewer dependents, which lowered the cost of living, but also meant that they could not send children to work. As a result, “married Black women were twice as likely to work for wages as other women”⁴¹ and over a third of all Black women had waged jobs. Additionally, by 1900, Black families had the highest proportion of boarders in the city – non-family members who earned their keep either through rent payments or through contributing subsistence labor to the family.

American born whites also depended on boarders, although to a lesser extent than other ethnic groups. They generally had children but pulled their children out of school to work at a lower rate than other ethnic groups. Stanley and Smith note that “this was possible in part because of the higher wages received by native White men, 72 percent of whom were either professional or skilled workers.”⁴² These families rarely sent married women into the workforce, although occasionally unmarried white women did work outside of the home.

³⁹Stanley and Smith, “The Detroit Story,” 37.

⁴⁰Stanley and Smith, “The Detroit Story,” 38.

⁴¹Stanley and Smith, “The Detroit Story,” 39.

⁴²Stanley and Smith, “The Detroit Story,” 38.

After the development of the Five-Dollar Day in 1914, household patterns in Detroit changed, but not necessarily in the direction of single-income nuclear families. Stanley and Smith note that in the decades following Ford's family-wage campaign, "even in Detroit, this presumed bastion of Fordism, the family wage was never a reality for the majority of wage-earners." Except for a small elite strata of the working classes, "the male wage was always supplemented by the wages of children and sometimes of wives, as well as by considerable amounts of subsistence labor."⁴³ This picture looks much the same as pre-Fordist Detroit, so what changed? Was the social transformation of Fordism as radical as its labor transformation?

In short, yes, but to understand exactly how we need to look more closely at patterns of race. Between 1900 and 1940, the population of Michigan more than doubled, from 2.4 million to 5.3 million. During this same period, the Black population of the state increased by a factor of more than *thirteen*, from just over 15,000 to more than 208, 000.⁴⁴ Most of this increase was driven by Black migrants from the southern states, looking for better working conditions and wages in the industrial Midwest.

To an extent they were able to find them, but they did not enter the Michigan workforce on equal footing with whites. According to the 1930 census, Black men were significantly overrepresented in low paid manufacturing jobs, while their white counterparts were more often employed either as shop foremen or skilled machinists.⁴⁵ The cities themselves mirrored this, with white workers more often owning free standing homes in newly developed neighborhoods, and Black workers and poor whites confined to the slums. In other words, the social significance of Fordism is defined as much by who it excluded from prosperity as by who it included.

Disparities in income are compensated for in many ways, and as Stanley and Smith

⁴³Stanley and Smith, "The Detroit Story," 34.

⁴⁴US Census Bureau. *Michigan*. 1940

⁴⁵US Census Bureau. *Michigan*. 1930

demonstrate, this is an essential element in labor history. It is not enough to note that Black laborers generally earned less than whites – we must look at the household forms that propped up this disparity and that allowed low-income groups to survive despite their condition of degraded labor. Stanley and Smith noted that Black families were much more likely to rely on boarders to make ends meet, and Black women were much more likely to work outside the home. Even within the “white” ethnic groups, there are various forms of family composition that contributed to the types of labor performed. Polish women laboring informally in vegetable gardens or caring for chickens are as much a part of labor history as American born unmarried white women working outside the home for a wage. And both are inextricable from the histories of the generally male, generally white worker that take up so much space in liberal and Marxist labor histories.

Global Complications: the Rubber Industry

Fordism did not develop in isolation, nor exclusively from local dynamics. Just as the existence of Fordist production methods was dependent on racial and gendered divisions in the local labor market, it was also built on a global division of labor, whereby the more brutal exploitation of workers in peripheral states propped up the productive relations in the core. Wallerstein terms this global dynamic the “axial division of labor,” a dynamic which ensures a “constant flow of surplus-value from the producers of peripheral products to the producers of core-like products.”⁴⁶

The nature of this global dynamic between the capitalist core and periphery was characterized by Andre Gunder Frank as the development of underdevelopment.⁴⁷ In his account,

⁴⁶Wallerstein, Immanuel. 2004. *World-Systems Analysis: An Introduction*. Durham: Duke University Press. 60

⁴⁷Andre Gunder Frank. 1966. “The Development of Underdevelopment.” *Monthly Review*. 18:4. 17-31.

the supposed economic backwardness of the global south is not a failure to keep pace with the capitalist core, but rather “is the result of its centuries-long participation in the process of world capitalist development.”⁴⁸ In other words, the underdevelopment of the global south is a condition positively created by the dynamics of global capitalism. Samir Amin’s *Accumulation on a World Scale* deals with this dynamic as well, attempting to counter the prevailing stagist narrative of underdevelopment which isolates the poverty of the periphery from world history and assumes that “laissez-faire was certain to develop the colonies just as it had developed the industrialized metropolitan countries.”⁴⁹ Amin insists that it is the capitalist core’s economic growth itself that creates the impoverished condition of the periphery. He claims that “as economic growth proceeds, none of these features by which the structure of the periphery is distinguished lessens; on the contrary, each increases.... In the periphery growth is *not* development, for its effect is to disarticulate.”⁵⁰

Economic disarticulation in the periphery is central to Amin’s analysis. In the core, national economies are largely integrated: different sectors “carry out substantial exchanges between themselves, which may be called “interindustrial” or “intersectoral” exchanges.”⁵¹ The outputs of extractive industries within developed economies directly support domestic manufacturing and agriculture. For so-called “underdeveloped” economies, internal trade between industrial sectors is stifled: “The different kinds of mineral wealth exploited by these great concerns - metals, oil, etc. - are not destined to supply domestic industries on the spot, but are exported in order to supply complex industrial groups in the advanced countries.”⁵² Whereas in the core, rationalization and growth in one industry is quickly generalized to others, in the

⁴⁸Frank. “The Development of Underdevelopment.” 21.

⁴⁹Amin, Samir. 1971. *Accumulation on a World Scale*. Monthly Review Press. 13

⁵⁰Amin, *Accumulation*, 18

⁵¹Amin, *Accumulation*, 16

⁵²Amin, *Accumulation*, 16

periphery industrial growth has minimal effect on the domestic economy as a whole, transferring the benefits to its industrial trading partners.

This dynamic can be traced along what world-systems analysts call commodity chains: the global chain of labor and raw-material inputs that results in the production of particular industrial and consumer goods. The term commodity chain was coined in a 1977 paper by Wallerstein and Hopkins, who propose it as an analytical tool to combat a “developmentalist” perspective in which “the world is said to consist of a number of related but basically autonomous ‘societies’ ... each moving upward along an essentially similar path of development.”⁵³ Criticizing an analytical nationalism which attempts to abstract autonomous societies from each other, Wallerstein and Hopkins insist on analyzing the world-system in toto: “a definite ‘world’, a spatio-temporal whole, whose spatial scope is coextensive with the elementary division of labor among its constituent regions or parts and whose temporal scope extends for as long as the elementary division of labor continually reproduces the ‘world’ as a social whole.”⁵⁴ Within this global social body, commodity chains form a sort of vascular system, transporting raw materials and ossified labor inputs from their sites of production, combining them, and bringing them to market where they can be valorized and turned into profit.

Commodity chains bring local and global dynamics into contact. As commodity production expanded over the course of capitalism’s global history, the need for vast amounts of raw materials to be transported across large distances increased. Bunker and Ciccantell claim that the “solutions to the raw materials problem require the coordination of multiple physical and social processes across geopolitical and physical space with domestic relations between firms, sectors, the state, labor, and new technologies.”⁵⁵ As a sort of vascular system for raw and

⁵³Hopkins, Terence K., Immanuel Wallerstein. 1977. “Patterns of Development of the Modern World-System.” *Review (Fernand Braudel Center)*. 111

⁵⁴Hopkins, Wallerstein, 1977, 112

⁵⁵Bunker, Stephen G., and Paul S. Ciccantell. 2005. *Globalization and the Race for Resources*. Baltimore: Johns

processed materials within the capitalist world-system, commodity chains are a primary axis of contestation for actors across ‘geopolitical and physical space.’ The state acts on commodity chains through trade agreements, or through nationalizing or privatizing reserves of raw materials. Labor asserts its power to strike or withhold itself at various points along the chain - labor inputs being the stuff that connects various links. And finally the firms, capitalists themselves, act to either subsume more and more of the production processes that feed into the final consumable as in Fordist vertical integration, or they outsource moments in the production process to locales with a cheaper labor supply as we see in the trends of post-Fordist globalization.

Commodity chains are a useful concept for countering overly-local views of capitalist accumulation. Analyses, such as Aglietta’s, that treat capitalist production in core regions as a self-contained whole fail to account for how productive relations in the core are propped up by quite different productive relations in the periphery. Smith, Ciccantell, and Sowers note that this problematic has been taken up at length by world-systems analysts who examine the racialized nature of labor in “resource frontiers and extractive peripheries.”⁵⁶ The effect of these “extractive peripheries on existing population, and especially indigenous populations, are almost universally negative, (see, e.g., Hall and Fenelon 2009; Gedicks 2001; Bodley 2014; Mann 2006), engendering often strong and violent opposition from indigenous groups (Ciccantell and Gellert 2022).”⁵⁷ The remainder of this section and the next will examine production and reproduction in this context along the Fordist commodity chain.

For early automobile producers in Michigan, many of these commodity chains were

entirely internal to the US. Many of the raw materials needed for the big three automakers to produce automobiles were readily available within the United States. Michigan's vast lumber and mineral resources were brought to Detroit by a robust rail system, and Ford even owned "mines in Kentucky and West Virginia, which gave him control over every natural resource needed to make a car - save rubber."⁵⁸ To acquire the rubber necessary for the production of automobiles, Ford and his competitors had to look outside the US - at first to Brazil and the Congo, and then later to Malaya.

Rubber has been known and used by Mesoamerican peoples for thousands of years - it was used to make bottles, balls or waterproof shoes. Despite this, in the 19th century the rubber industry in Europe and the US mainly provided for products with limited industrial practicality until the development of the vulcanization process in the late 1830s. Vulcanization allowed for the creation of stable and hardened rubber industrial products, and by 1945 carriage wheels were equipped with solid rubber tires.⁵⁹ Tire production, along with rubber insulation for electrical wires, drove the rubber industry to new heights. In the US, the rubber industry was centered in Ohio - by 1900 Akron was the home to Goodrich, Goodyear, and Firestone. Business was good and all three were multi-million dollar companies by 1910. Firestone was especially successful, thanks to exclusive contracts with Ford to provide tires to the Model N in 1906 and to the famous Model T in 1908.⁶⁰ Fortunes were made in the manufacture and sale of tires, but down the commodity chain the story was quite different.

Industrial production of rubber began in Brazil and central Africa in the 1800s, where latex producing plants grew natively. Bunker and Ciccantell claim that since "the Amazon provided the technically most useful and abundant kinds of rubber in the world, states and firms

⁵⁸Grandin, Greg. 2013. "Empire's Ruins: Detroit to the Amazon." in *Imperial Debris*, ed. Ann Laura Stoler. Durham: Duke University Press. 118

⁵⁹Coates, Austin. 1987. *Commerce in Rubber: The First 250 Years*. Oxford: Oxford University Press. 49

⁶⁰Coates, *Commerce in Rubber*, 134

in the rapidly industrializing, trade-dominant nations of Europe and America were particularly interested and active in the Amazon.”⁶¹ As in the Congo, initial production consisted of wild-collection by contract laborers. Rubber plants were quickly exported to tropical regions around the globe such as southeast Asia, although there was little commercial production in Asia until the collapse of the rubber market in the Congo and the Amazon in 1913.⁶² Although plantations would dominate the global rubber production by the second decade of the twentieth century, for the first several decades of the auto industry there were few rationalized rubber plantations, and rubber had to be sourced from wild collectors in the Amazon and the Congo. By the end of the nineteenth century, the Amazon was supplying the world market with 15,000 tons of rubber annually.

The working conditions in rubber fields in Brazil were a world away from those in the US. In the late 1870s a severe drought struck Northeastern Brazil, destroying the means of subsistence for a large portion of the population. Many left their homes and found work in the growing Brazilian rubber industry, but under the harshest of conditions. Forced to pay for their own provisions on the way to the work locations, rubber tappers started working in severe debt. The rubber companies held this debt on account against future rubber yields, but “every time [the worker’s] rubber (his credit) was collected, it was found that his debt had mounted higher than his earnings.”⁶³ The companies ensured that the prices of necessary provisions were higher than the piece-wages for rubber, creating a vicious cycle of debt peonage.

The Brazilian rubber industry was quite volatile, in large part because of the inconsistent nature of the production system. Rubber in Brazil was not produced on plantations, but was gathered from wild rubber trees by contract workers. Although the exploitation of tappers by

⁶¹Bunker and Ciccantell, *Globalization*, 39

⁶²Coates, *Commerce in Rubber*, 115

⁶³Coates, *Commerce in Rubber*, 94

traders and barons was significant and the social arrangement was inefficient, there were several factors that impeded the rationalization and industrialization of rubber production in Brazil.

Oliver T. Coomes and Bradford L. Barham note two explanations for the relative failure of proletarianization among Brazilian rubber tappers.⁶⁴ The first was tapper resistance: the desire among tappers to maintain the relative autonomy of piece work and their aversion to wage-labor. Along with the traders, tappers actively resisted the development of plantations. The second explanation was “a South American leaf blight, which flourished in closed-canopy Amazonian rubber stands”⁶⁵ that damaged the Amazonian crop relative to rubber grown in Asia.

Although rubber production in Brazil looked very different from the Fordist production in America, it was in no way independent of industrial developments in the global core. The demand for rubber created the incentive for rubber barons to squeeze as much labor from an impoverished workforce. The inhospitableness of the Brazilian rubber industry to plantation-style cultivation did not stop American magnates from trying to mold the industry in their own image. However, they had very little success. In the 1920s, Henry Ford attempted to establish a rubber plantation in Brazil based on the production design principles of his auto factories. This foray into rubber production, a plantation town called Fordlandia, was an unmitigated disaster.

Ford’s brilliance lay in his understanding of how the social life of workers can affect the productivity of a firm. His Five-Dollar-Day was a true innovation in productive relations. But though he understood that the articulation between production and reproduction was important, he misunderstood its true nature. His planned company town, Fordlandia, was an attempt to transplant the productive/reproductive dynamic that was so successful in Michigan to the rubber industry in Brazil. Because it failed to take into account local productive, social, and natural

⁶⁴Coomes, Oliver T. and Bradford L. Barham. 1994. “The Amazon Rubber Boom: Labor Control, Resistance, and Failed Plantation Development Revisited.” *Hispanic American Historical Review*. Vol 74:2

⁶⁵Coomes and Barham. “The Amazon Rubber Boom”

conditions, Fordlandia was never viable.

In early 1928, Henry Ford announced that he had acquired a sizeable tract of land in the Amazon, and that he planned to establish a rubber plantation, firmly bringing the automobile's last natural input under Ford's control. The announcement was met with acclaim - at last the forces of rationalization and civilization would meet and conquer the wild territory of South America. Ford's plantation would take hold in the jungle and revolutionize the entirety of the inefficient rubber industry - newspapers proclaimed Ford's project of cultivating not only "rubber, but the rubber gatherers as well."⁶⁶

Greg Grandin describes Ford's project as a transplantation of both community and production patterns. Ford "tasked his managers with building Cape Cod-style shingled houses for the Brazilian workforce he hired. He urged the workers to tend flower and vegetable gardens, and to eat whole wheat bread, unpolished rice, canned Michigan peaches, and oatmeal."⁶⁷ Disregarding the extant conditions and patterns of life among rubber workers, he tried also to impose a prohibition on alcohol, and "Fordlandia had a central square, sidewalks, indoor plumbing, manicured lawns, a movie theater, shoe stores, ice cream and perfume shops, swimming pools, tennis courts, a golf course, and Model Ts rolling down its paved streets."⁶⁸ The field, too, he tried to turn into a shop floor. Grandin notes that Ford "insisted that his managers plant rubber trees in tight rows - back in his Detroit factories, Ford famously crowded machines close together to minimize movement."⁶⁹

In both cases, these attempts to extend patterns of rationalization unchanged from the core to the periphery backfired. Workers resisted Ford's puritanism and the rigidity of suburban life - Grandin notes that "Fordlandia in its early years was plagued by vice, knife fights, and

⁶⁶Washington Post, cited in Grandin, "Empire's Ruins," 118

⁶⁷Grandin, "Empire's Ruins," 118

⁶⁸Grandin, "Empire's Ruins," 119

⁶⁹Grandin, "Empire's Ruins," 119

riots. The place seemed less *Our Town* than *Deadwood*, as brothels and bars sprawled around its edges.”⁷⁰ The industrial policy was just as misplaced as the attempt at social engineering.

Workers resisted also the rigidity of production and the relative lack of freedom that plantation work implied. The plantations themselves, modeled on the tight spaces of Michigan’s factories, were designed without consideration of the unique biodiversity in the Amazon. By planting monocrop fields of rubber trees, Ford “actually created the conditions for the explosive growth of the bugs and blight that feed off rubber, and these eventually laid waste to the plantation.”⁷¹ In the roughly two decades of its existence, Fordlandia failed to produce any rubber that ended up in a Ford car.

Ford planned for his plantation project to supersede the patterns of wild rubber cultivation that had prevailed until the mid 1910’s. It represented an attempt at a less violent form of labor control than that which existed in the Amazon and the Congo, as well as an attempt to export the principles of rationalized production. In both attempts, Ford failed. Although rationalized plantation production failed in Fordlandia, where it was combined with Michigan-style reproductive conditions, it thrived in the conditions that prevailed in British Malaya. In 1913 the wild rubber market in the Amazon and the Congo collapsed, and a new world market for rubber developed in southeast Asia, centered on British Malaya⁷² where plantations operated with indentured South Indian migrant labor. This system “had always been a disreputable business, an exploitation of the poverty-stricken for whom there were no guarantees,”⁷³ but it replaced some of the more brutal methods of labor control in wild rubber collection without providing the security offered to Michigan workers up the commodity chain. The next section will look at the living conditions of auto workers in the US, and then will revisit workers in

⁷⁰Grandin, “Empire’s Ruins,” 119

⁷¹Grandin, “Empire’s Ruins,” 119

⁷²Coates, *Commerce in Rubber*, 115

⁷³Coates, *Commerce in Rubber*, 110

rubber fields in the Amazon, the Congo and on Malayan plantations. On that basis we will work towards a framework of social reproductive dynamics that extends from the core to the periphery.

Housing and Social Reproduction along the Fordist Commodity Chain

The lesson of Fordlandia is not that the Brazilian Jungle was too wild to be tamed, or that its workers were somehow unready for rationalized proletarianization. Rather, the failed dynamic of peripheral/core relations within the Fordist outpost highlights what Gunder Frank called the development of underdevelopment. The strictures of the world-system did not allow for the generalization of production conditions in the core, but required the development of severe conditions specific to the periphery. The articulation between core-like productive relations and peripheral productive relations is of central concern to world-systems analysts, and mapping its contours in the Fordist case is the first step in my project. My ultimate goal is to situate housing policy and the social-reproduction of the labor force within this global dynamic. How do social reproductive pressures manifest at various points along the global commodity chain, and how do structures of housing respond to and influence them?

We are finally at a point to revisit the role of housing in labor history generally, and in the rise of Fordism specifically. This task is made difficult by the inconsistent collection of meanings assigned to the word 'Housing'. However, I think that they can be fairly easily grouped into three rough (but overlapping) categories: the physical attributes of housing, its geographic organization, and its position in the social relations that make up society. This section will sketch out some preliminary insights into the relationship between housing and labor and will lay out a plan for future study.

At its most basic, ‘housing’ refers simply to the physical configuration of stones, wood, and steel that we prop up to keep ourselves safe from the wind and rain. A study of housing could justifiably include an analysis of building materials, and the shapes we assemble them into: ranch houses, apartment complexes, townhomes, or even barracks. What is physically included in the house is relevant and changes over time: the innovation of indoor plumbing is significant, and the position of the kitchen as either a free-standing structure or as a part of the living space.

At first glance, the relevance of Michigan’s pre-Fordist industry in the 19th century to the construction of *physical houses* is obvious: Michigan’s economy was based in logging, and the lumber that was cut down was largely used to provide framing for houses in the growing cities, and in the treeless Great Plains states. But the relationship was two-sided. The lumber industry largely existed because of a need to build houses, but the nature of lumber work itself also influenced the organization of workers’ housing. The mobile nature of the lumber industry meant that workers had to live in cheaply constructed barracks, rather than single-story homes or apartments fit for a family. Instead of individual kitchens for each inhabitant, the camps had large cookhouses that could easily prepare three meals a day for a whole company of men.

It is difficult to separate the physical construction of housing from its geographic organization. Indeed, as the lumber industry moved west from Michigan and the state turned to manufacturing, workers left the barracks behind and moved to cities. Between 1890 and 1920 the population of Michigan grew from 2.1 million to 3.7 million: an increase of over 75%⁷⁴. In the same period Michigan’s three largest cities, Detroit, Flint, and Grand Rapids, grew by a combined 946,954 – Flint and Detroit each growing more than fourfold.

Much of this growth was made up of immigrants from central and eastern Europe who migrated as families rather than as single workers. Dunbar notes that urbanization in Michigan

⁷⁴US Census Bureau. *Michigan*. 1920

outpaced its neighboring states in the first decades of the 20th century. Immigrants largely came to the cities in search of stable manufacturing jobs. However, this growth posed significant challenges to the cities. Since the needs of working families in urban areas could not be met by roving barracks and cookhouses, or the institutions of rural life, “every city had to cope with crime and delinquency, public-health problems, the need for recreational facilities, and dozens of other problems.”⁷⁵ The growing need for skilled labor was also mirrored in a need for a more robust education system.

Running through the physical manifestations of housing is an endless series of social relations. These are manifold, ranging from property relations, to housing codes and policy, to racial segregation and urban policy, to household structure and family forms, all influenced by the state. Through legislation, law enforcement, and funding the state is able to directly intervene into the social structure of housing. A 2004 report on the history of housing legislation published by the Congressional Research Service notes the earliest objects of modern housing legislation: the purchase and improvement of slum properties by the state, and the provision of housing for shipyard employees and war workers during the first World War.⁷⁶

However, the federal government would not systematically deal with housing welfare until the Fair Housing Act of 1934 and then more thoroughly with the Department of Housing and Urban Development in 1965. Until then state intervention into housing mostly existed at the local or state level. Bolstered by the 1896 Supreme Court case *Plessy vs Ferguson*, which established *de jure* segregation through the principle of ‘separate but equal’ facilities for white and black citizens, the local governments and militant Jim Crow segregationists mostly intervened in housing to enforce segregated cities. According to Carl H. Nightingale’s study on

⁷⁵Dunbar, *Michigan: A History of the Wolverine State*, 582.

⁷⁶Congressional Research Committee. 2004. *A Chronology of Housing Legislation and Selected Executive Actions, 1892 – 2003*. Washington: U.S. Government Publishing Office.

segregation, these groups “aimed to convince state and federal authorities, including the judiciary, to extend Jim Crow to residential neighborhoods in cities across the country.”⁷⁷

Jim Crow activists had powerful allies with much to gain from racial segregation: the industrialists. We have noted above that the automotive industry benefited from a stratified workforce with a depressed and racialized underclass. But the auto companies did not merely benefit from a pre-existing situation: they actively worked to create it. Manipulation of the housing market was a primary tool they employed in the creation and maintenance of this stratified market.

Henry Ford’s Five Dollar Day indirectly encouraged residential segregation by withholding the increased pay from workers who lived in “bad” or neighborhoods with “foreign” neighbors.⁷⁸ However, General Motors explicitly manipulated housing in order to maintain a stratified labor market. In 1919, General Motors created a housing division under the name the Modern Housing Corporation. The primary goal of this division was to directly provide housing for employees to combat Flint’s housing shortage. Rather than follow the company-town model of places like Pullman, Chicago or the mining camps of Michigan’s Upper Peninsula, GM purchased land, constructed houses along with amenities such as schools, parks, and sidewalks, and then sold the houses to employees.⁷⁹

This project created a class of homeownership employees with full access to public amenities. However, it followed a policy of racial segregation – by refusing to sell homes to Black employees, it consigned them to the slums and poorer neighborhoods of Flint. Here, as the cost of living (as well as the quality of life) was lower, so also was the cost of their labor.

⁷⁷Carl H. Nightingale, 2012. *Segregation: A Global History of Divided Cities*. Chicago: University of Chicago Press, 2012. 303

⁷⁸Stephen Meyer III. 1981. *The Five Dollar Day: Labor Management and Social Control in the Ford Motor Company, 1908 – 1921*. Albany: State University of New York Press. 133

⁷⁹Cowdrick. 1930. “Operations of Modern Housing Corporation.” General Motors Library. Accessed Dec 7, 2021.

Little has been written about the Modern Housing Corporation, but I believe it to be essential in bridging the gap between the pre-Fordist housing configurations of work camps, or company-owned mining towns, and the welfare housing developed by the state during the Great Depression of the 1930s. General Motors, along with the Ford Motor Company, saw the organization of housing as directly relevant to their bottom line. They manipulated housing structures to create the ideal of a modern home-owner, but also to control the variable costs of different types of labor. As the economy entered the crisis of the 1930s, the state took over responsibility for maintaining this aspect of social reproduction.

Housing remains a crucial element of Fordism even when we look beyond the boundaries of the industrial core, albeit it looks very different. Henry Ford's attempt to control rubber production was not just a reconfiguration of the production processes, but of the articulation between production and consumption. The development of Fordlandia as a Michigan-style city in Brazil was a naive attempt to transport core-like social relations to the periphery, but its failure does not indicate that Fordist production was not a determining force in the social reproduction of labor in the south. Rather it indicates that Ford's influence was filtered through the relations of the world-system, and that it manifested in the more direct forms of social and labor discipline in the rubber producing regions of the Congo and Latin America.

The two patterns of rubber production - wild collection and planned cultivation - were sustained by different patterns of social reproduction. Wild rubber harvesting was predominant in the Amazon and the Congo, and dominated the world market until 1913.⁸⁰ In Brazil, before the founding of Fordlandia, rubber cultivators operated on a pseudo piece-work basis. The costs of production were borne by the worker, who ventured into the jungle to collect raw rubber that he would then sell to the company. This work was seasonal, and depended on a transhumant

⁸⁰Coates, *Commerce in Rubber*, 162

workforce that traveled between port towns and rubber estates as weather conditions dictated.⁸¹ A stable workforce was neither required nor desired in Brazil, and so there was no impetus for the companies to develop stable reproductive relations for workers. Rather workers were compelled to work through the cultivation of debt - through the sale of “the necessary working tools and basic goods such as foodstuff, which were forcibly acquired at the rubber patron’s storehouse or station.”⁸² A continual supply of labor was ensured through the debt contract.

The labor supply in the Congo was ensured through similar means, but this time with a tax burden rather than a debt burden. In a comparative study of Amazonian and Congolese rubber extraction, de Mello and Van Melkebeke note that from the 1890s on, an “in kind rubber tax ... proved to be the solution to increase revenue.” They go on to say that “this huge increase in exports received by in-kind taxation would result in severe and violent labour mobilisations and would cause devastation throughout the Congolese countryside.”⁸³ Whereas in the Amazon, workers migrated to the rubber regions due to harsh economic and environmental conditions, in the Congo the state and private firms sent expeditions into the countryside to capture and enslave native men. “Expeditions often ended up in plunder, destruction of villages and capturing of people: initially, to collect ivory (by the Congo Free State) and subsequently, to collect rubber (by the Congo Free State and private sector).”⁸⁴ These workers were then forced to collect and deliver “six to eight kilograms per person every two weeks” and “when they did receive some sort of remuneration, this was usually in-kind (salt, knives, beads, textiles, etc.).”⁸⁵

The relevance of housing in the Amazon and the Congo lies in the *absence* of any dedicated housing policy. If the impact of Fordism on housing in the core was to expand the

⁸¹Coomes and Barham, “The Amazon Rubber Boom”. 242

⁸²de Mello, Louise Cardoso and Sven Van Melkebeke. 2019. “From the Amazon to the Congo Valley,” in *Commodity Frontiers and Global Capitalist Expansion*. Ed. Sabrina Joseph. Palgrave MacMillan. 148

⁸³de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 143

⁸⁴de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 147

⁸⁵de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 152,3

commodification and the housing market, its impact in the periphery was the opposite. Existing communities and their social relations were violently disrupted by rubber barons and the state who saw no need to recreate conditions for the renewal of laborers. Rather indigenous and migrant populations provided an external pool from which new labor could be drawn. The pay given to rubber workers (if it can be called pay) was insufficient to provide for the renewal of the conditions of labor on an open commodity market, and so the gap was filled by coercion, and by the workers' informal reproductive labor. Although the workers were undoubtedly commodity producers, they were not commodity consumers in the same sense that workers in the core were. Their everyday necessities were obtained from closed company shops, and in both the Amazon and the Congo, roving rubber tappers built their own temporary shelters in the rubber fields.⁸⁶ When debt and tax burdens were insufficient to compel labor in these conditions, violence was used, and when violence was insufficient workers were killed, and new workers brought in.

We see that in the wild rubber fields, gendered labor dynamics differed significantly from those in industrial Michigan. Many Michigan women performed unpaid reproductive labor to support and supplement their husband's 'family wages' and in the poorer communities they took on piecework or service work in addition to this unpaid care work. In the core, this gendered dynamic was structured in part by the housing policies of the industries, but in the periphery the gendered division of labor manifested quite differently. Maria Mies' analysis of women and colonialism helps us contextualize the dynamics of gendered labor in the periphery. Writing about the 18th and 19th centuries, she notes that where unfree labor was secured through slaving raids and trade, "slave women were actively discouraged from bearing children or forming families."⁸⁷ As long as the slave trade was thriving and the problem of reproducing the labor

⁸⁶de Mello and Melkebeke, "From the Amazon to the Congo Valley," 153

⁸⁷Mies, Maria. 1986. *Patriarchy and Accumulation on a World Scale*. London: Zed Books. 91

force was resolved through external means, enslaved women were expected to perform hard labor alongside men. When the atlantic slave trade was abolished, this dynamic shifted, and enslaved women were forced to bear and raise children - future workers. Mies notes that colonial administrators enacted “incentives in the slave codes of the late eighteenth and nineteenth centuries to encourage local breeding of slaves by slave women on the plantations.”⁸⁸

How did these dynamics play out in the twentieth century in the rubber fields? de Mello and Melkebeke seem to indicate that the 19th century practices of slaving raids in the Congo, in which “people were captured and enslaved, usually men, but also women, who were incorporated into the raiders’ harems,” were continued by rubber producers in the 20th century.⁸⁹ However, women also occasionally worked in the rubber fields. As a result of this system, the Congo witnessed a significant demographic decline, with “contemporary observers and present-day scholars [agreeing] on rural depopulation and decimation of the Congolese,” along with “numerous cases of depopulated villages”⁹⁰ - homes standing empty save for ghosts. The situation in the Amazon was analogous: “In general terms, the demographic post-boom scenario is marked by the emptying of villages or whole regions, being caused either by workforce recruitment, the resulting unsustainable livelihood, death by diseases, ethnocide, demographic replacement and miscegenation or encroachment.”⁹¹

As wild harvested rubber declined on the world stage, hampered by the South American blight in the Amazon, King Leopold’s personal finances in the Congo, and the inefficiency of production everywhere, it was replaced by plantation rubber in southeast Asia. Rubber plants were first brought to Malaya in the 1870s, but were not cultivated on a large scale until 1897-8

⁸⁸Mies, Maria. *Patriarchy and Accumulation*. 91

⁸⁹de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 145,6

⁹⁰de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 160

⁹¹de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 161

when plantation owners were looking for a replacement for failing coffee crops.⁹² With the Malay Reservations Enactment in 1913, foreigners were allowed to purchase land in certain areas and large rubber plantations developed, requiring a large immigrant workforce. To supply this demand, the government of the Federated Malay States proposed a policy centered around three points: the development of a free workforce; the recruitment of immigrant workers from India, and regulation of employment conditions (e.g. hospitals, water, housing etc.).⁹³

In practice, this policy led to significantly better working conditions than those in the Amazon and the Congo, but worse than those of auto workers in Michigan. J. Norman Parker, in *Colonial Labor Policy*, notes that in Malaya, “to be a free laborer often had little real meaning since the working conditions of a free laborer and an indentured laborer differed little or not at all.”⁹⁴ Most of the workers, indentured or free, worked for a wage that was differentiated on type of job and the sex and race of the worker. Women generally received “between 70 and 80 percent of male laborers’ wages”⁹⁵ and Chinese laborers received more than Indian or Javanese workers. For all workers, the workday was limited, although planters struggled to extend it.

In Malaya, a steady supply of workers was ensured by the enticement of good pay, and “in the possibility of his being able, during his two or three years average stay in this country, to remit small amounts of money to relations.”⁹⁶ As in the Congo and the Amazon, the labor force was not reproduced locally and generationally, but through tapping an external source of potential workers. In this case, the government of India worked with Malaya to facilitate the migration of workers from southern India to the Malayan rubber fields. Unlike in the wild cultivation fields across the world, workers in Malaya were paid in cash, and employers were

⁹²Drabble, J. H. 1967. “The Plantation Rubber Industry in Malaya up to 1922.” *Journal of the Malaysian Branch of the Royal Asiatic Society*. Vol 40:1. 54

⁹³Drabble, “Plantation Rubber Industry,” 59

⁹⁴Parmer, *Colonial Labor Policy*, 16

⁹⁵Parmer, *Colonial Labor Policy*, 168-9

⁹⁶Parmer, *Colonial Labor Policy*, 177

directed to provide for workers' housing, sanitation and medical care in addition to wages.⁹⁷ In the 1920s, the level of acceptable wage and quality of social provisions was a fixed point of contention between the government of India, who facilitated the supply of immigrant workers, and the government of Malaya and the planters, who wanted to reduce costs.

The question of social reproduction is complicated by the introduction of migrant and indentured labor. As with seasonal/transhumant labor in the Amazon, the social reproductive dynamics of migrant workers in colonial Malaya were stretched across vast distances. The existing communities of workers in India and Malaya were disrupted by migration, but do not seem to have been destroyed as in the Congo and among indigenous Amazonians. Rather, the social reproduction of the workforce became bifurcated, operating locally within the amenities of the rubber plantations and regionally through cash transfers across borders. This introduces significant complications to the question of housing and labor reproduction, but provides an exciting opportunity to examine the fundamental relationship between labor and housing in capitalism as it is attenuated by geographical and political space.

A steady supply of laborers is a central problem for industry everywhere. The housing and social-reproductive conditions for laborers varied along the global automotive commodity chain. Even at the point of the chain where rubber entered the production process, the living conditions of workers in different regions were structured differently by global and local pressures. The conditions of the reproduction of labor responded to factors such as the availability of cheap labor from an outside source, the resistance of workers to changes in their social and communal patterns, and the type of labor required by capital .

Conclusion: An investigation into Housing and Labor

⁹⁷Parmer, *Colonial Labor Policy*, 177-8

Social Reproduction Theory and World-Systems Analysis provide robust theoretical frameworks for conceptualizing the manipulation of labor and its value across racial and familial axes. Both will be helpful in diagnosing the manipulation by capital of the social relations that run through housing. However, neither uses ‘housing’ as a conceptual or analytic category, rather employing the related but distinct categories of ‘family’ and ‘the household’. To this end, the necessary empirical investigation into the housing policies and structures of Fordist Michigan and the global south will need to be paired with an analysis of how these theoretical categories are related and co-constituted.

The primary question posed by this paper is that of the articulation between capital’s need for specific types of labor, and the specific ways in which it reproduces these types. In particular, how did global configurations of social reproduction respond to the Fordist revolution in production? Following from the insights of Marxist Feminists and social reproduction theorists, I posit that housing conditions structure the reproduction of labor, but this is not a simple position to investigate. It is clear that neither conditions of production nor reproduction were exported globally wholesale by Fordist producers, and yet the entire world of production and reproduction was certainly transformed.

In the Amazon and the Congo, the reproduction of the debt relation and of the tax relation is inextricable from the reproduction of labor. The housing policy in these fields was a negative one: a policy of destroying existing housing patterns. The razing of homes by slaving expeditions in the Congo, and the “kidnapping of indigenous infants ... to be raised within the families of enforcers or patrons to work the rubber fields”⁹⁸ in the Amazon are the logical parallels of the planned communities in Michigan. In Michigan a labor supply was ensured and renewed through the new family patterns of the middle class; in the global south this reproduction of labor was

⁹⁸de Mello and Melkebeke, “From the Amazon to the Congo Valley,” 151

ensured and renewed through the destruction of indigenous families.

The supply of labor on the Malayan plantations was also secured from external sources, but through economic enticement rather than brute force. Plantation workers were paid a wage, but as their housing was provided for by the planters, and the state enforced labor laws regarding the length of the workday and workweek. The tenure of migrant workers was short, and for many workers a portion of his/her wage was used to support families back home in India.

These patterns of reproduction are connected across the world-system by commodity chains. A finished car needed inputs of iron, wood, coal, and rubber, and each of these had to be gathered and prepared by laborers far from auto factories. The labor processes, and the conditions of its renewal, were conditioned by various local factors - the position in the global economy, the extant patterns of life and labor, and the family formation in which laborers lived. While factory workers in America could be said to be proletarians, slave laborers and informal workers in the rubber fields could not. And while production for the world market dominated the lives of workers and their families, it was not the only type of work they engaged in: they were supported by unique combinations of wage labor, unpaid reproductive work, and subsistence labor.

The ways in which workers across the automotive commodity chain were housed, or the ways in which they housed themselves, are constitutive of the capitalist world-system. Aglietta was correct in identifying housing as a key element in Fordism, but he was incorrect in dismissing the relevance of the global south to Ford's project. If Fordism can be seen as a reconfiguration of the articulation between patterns of consumption and production, then the changes that we have traced across the global south form a part of the regime of Fordism and warrant further study within this framework.

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